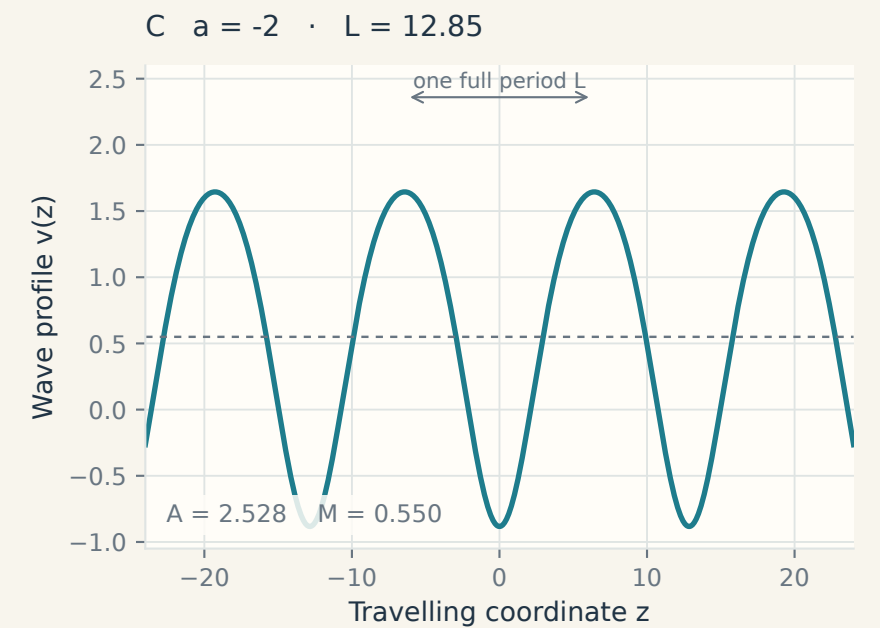
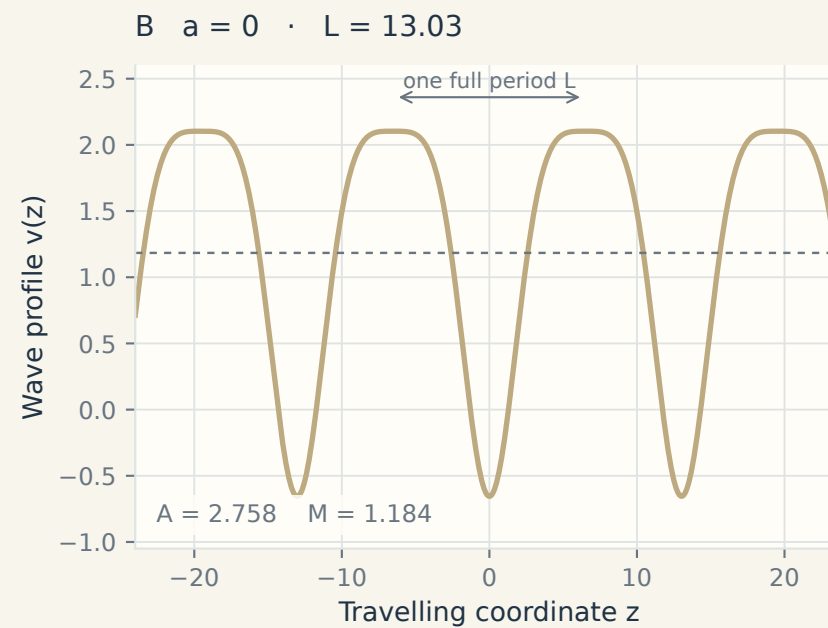
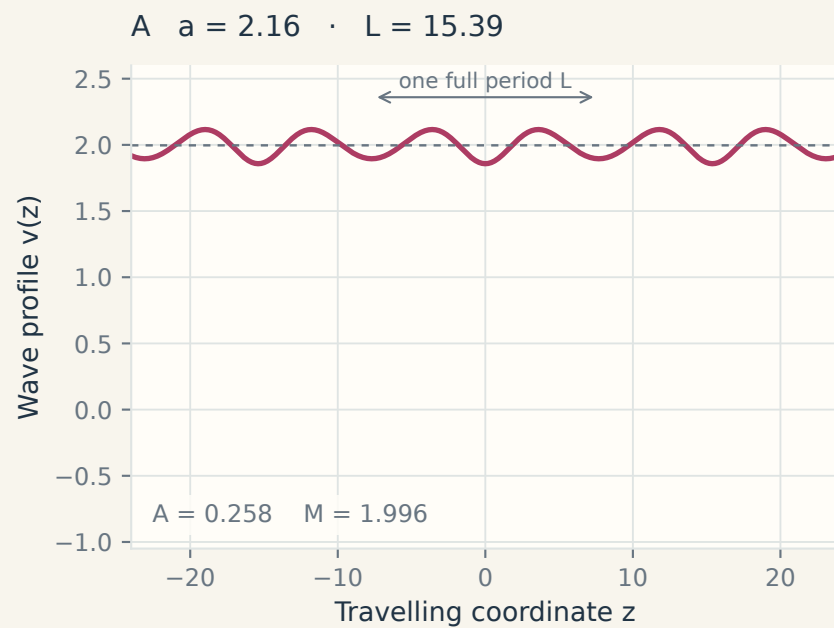
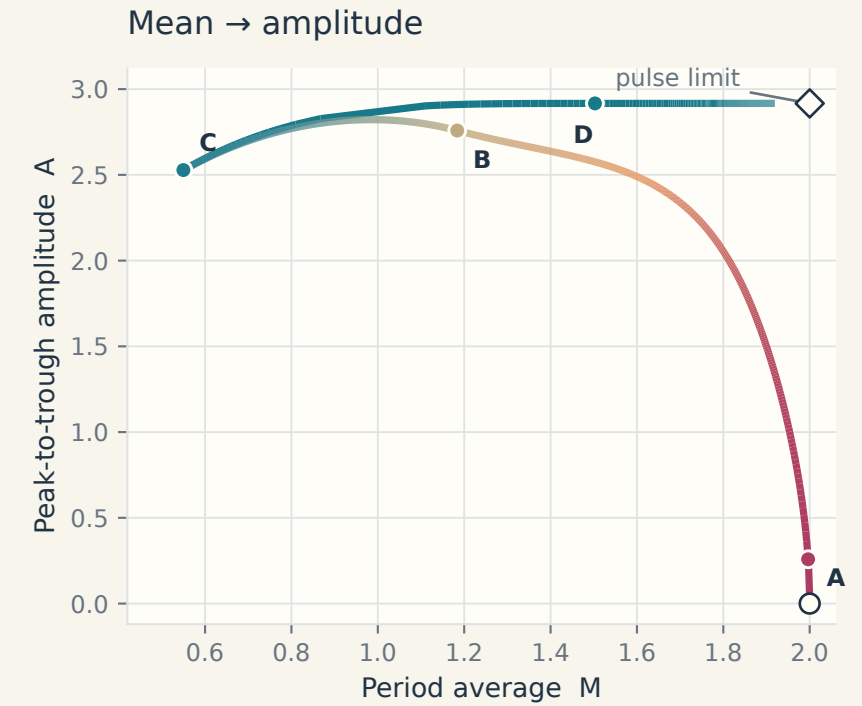
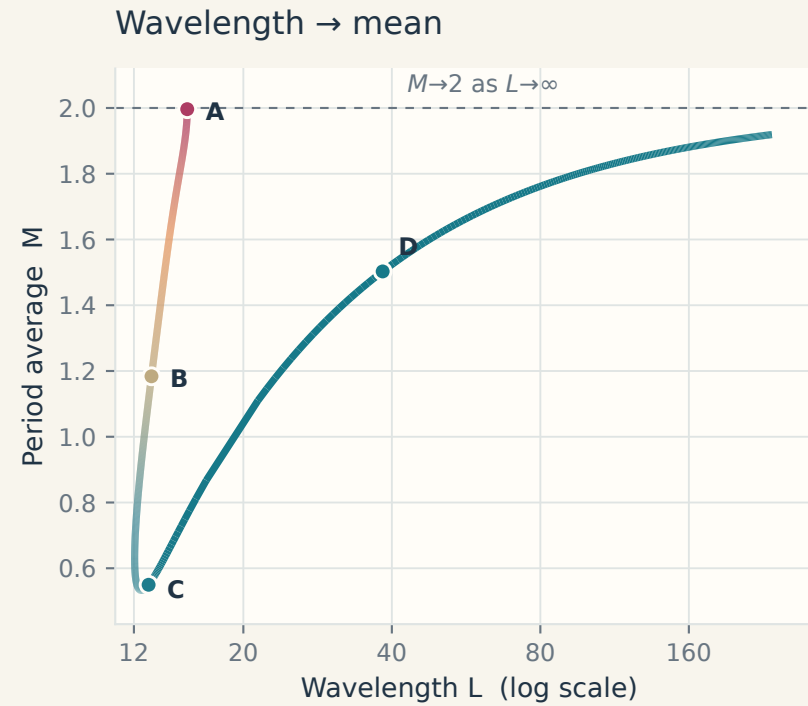
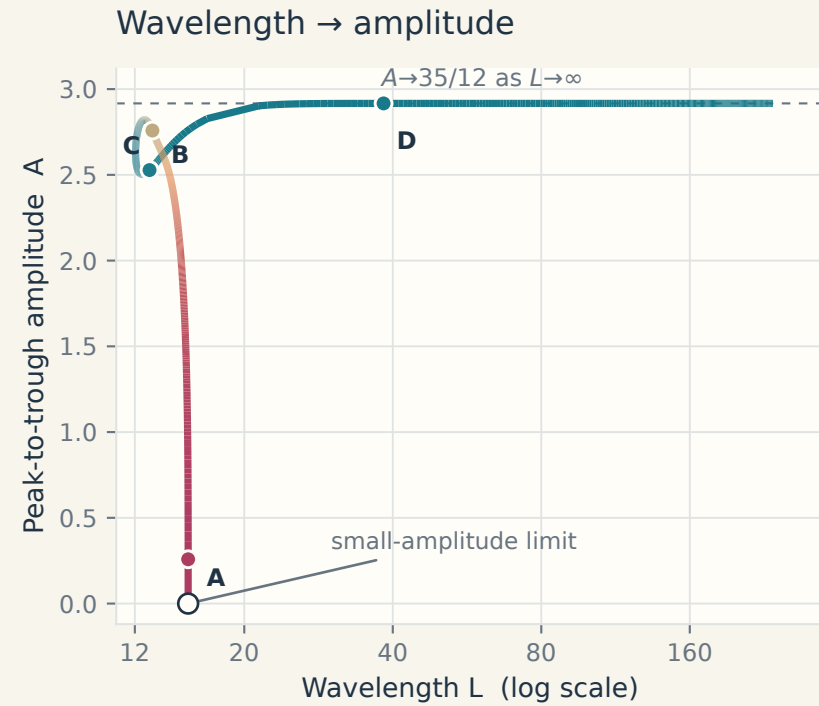


Kawahara in physical coordinates

$$v'''' + av'' - v + v^2/2 = 0 \quad \cdot \quad \text{smooth real periodic branch} \quad \cdot \quad -13/6 < a < 13/6$$



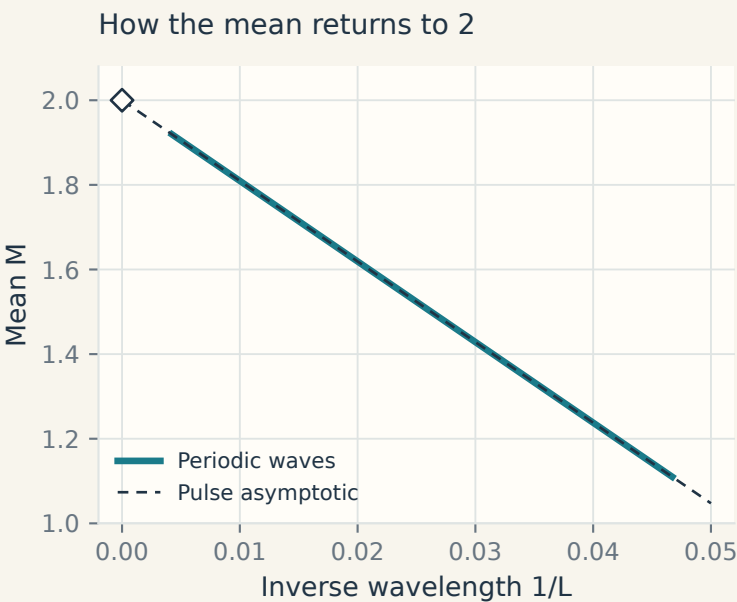
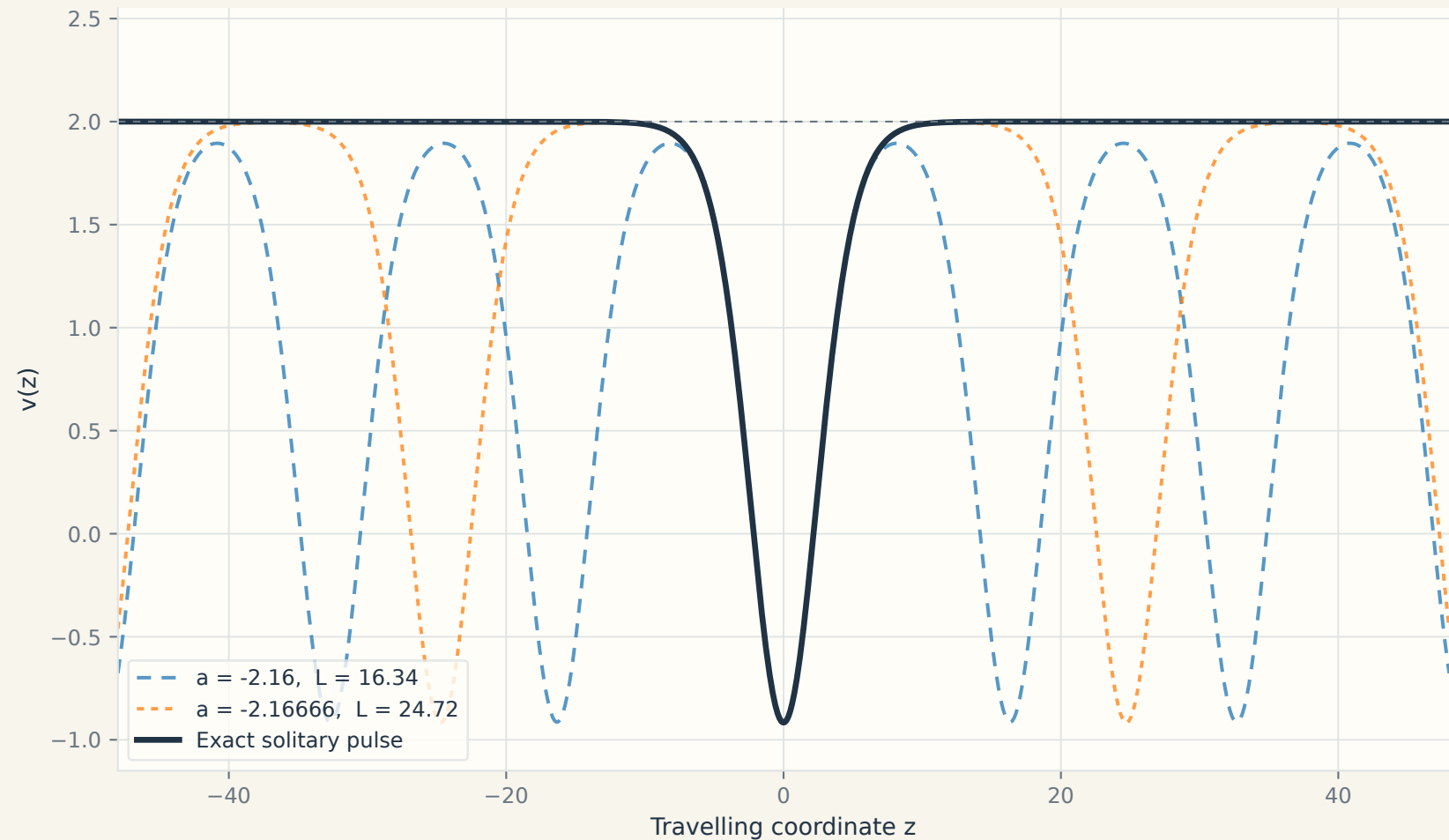
Dots A-D identify the same waves in each projection.
D is a long-period wave ($L = 38.32$).
Dashed profile lines show the period averages.

L is the full repeat distance; $A = \max(v) - \min(v)$; M is the average over one period. All units are those of the displayed normalized equation.

The pulse has infinite wavelength and is shown only as a limit. At $a = 13/6$ the profile becomes constant, so its wavelength is also a branch limit.

A periodic train opens into a solitary pulse

$a \rightarrow -13/6: L \rightarrow \infty, A \rightarrow 35/12, M \rightarrow 2$

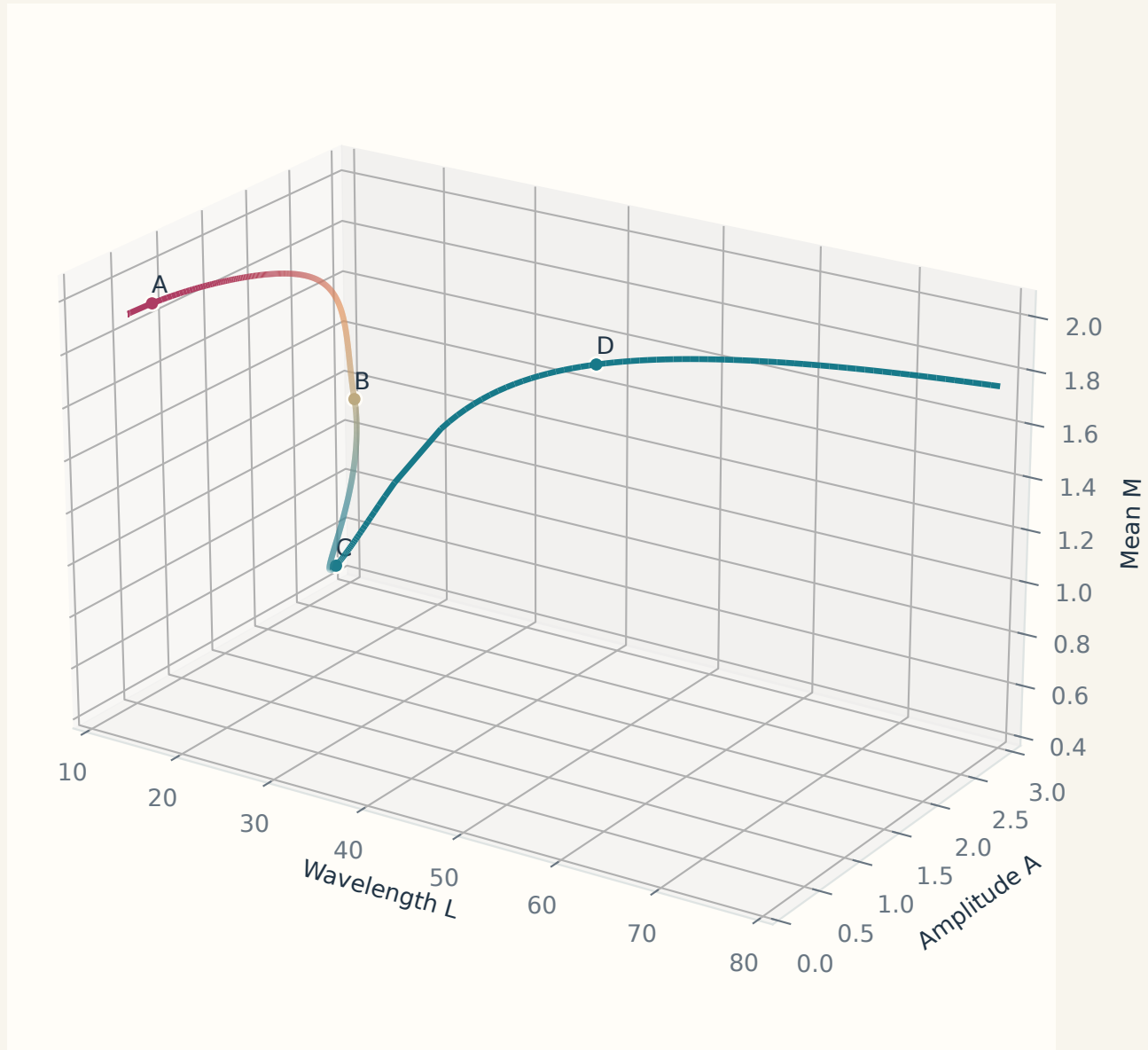


$$v_*(z) = 2 - \frac{35}{12} \operatorname{sech}^4\left(\frac{z}{2\sqrt{6}}\right)$$

$$M = 2 - \frac{70\sqrt{6}}{9L} + o(L^{-1})$$

The pulse keeps a finite depth while the separation between pulses diverges. Its mean approaches the background because a single depression occupies a vanishing fraction of the period. The full ODE identity, mean formula, extrema, and this limiting profile are checked by the accompanying script.

One branch in (wavelength, amplitude, mean)



Color follows a. The long-period tail continues beyond the window toward $A = 35/12$ and $M = 2$.